

Disabling Pansclerotic Morphea of Childhood (DPM): A Comprehensive Disease Characteristics Report

Disease: Disabling Pansclerotic Morphea of Childhood (DPM / DPMC) **MONDO ID:** MONDO:0957497 · **OMIM:** #620443 · **Causal gene:** *STAT4* (gain-of-function) **Category:** Mendelian (autosomal dominant) autoinflammatory fibrosing disease

Summary

Disabling Pansclerotic Morphea of Childhood (DPM) is an ultra-rare, childhood-onset fibrosing disorder historically regarded as the most severe end of the juvenile localized scleroderma (JLS)/morphea spectrum. Until recently it was considered an idiopathic/autoimmune condition of unknown cause. In 2023, a landmark study by Baghdassarian and colleagues (*New England Journal of Medicine*) reclassified DPM as a **Mendelian, autosomal dominant, autoinflammatory disease** caused by heterozygous **gain-of-function (GOF) missense variants in the DNA-binding domain (DBD) of *STAT4*** ([PMID: 37256972](#)). Three novel variants — p.His623Tyr, p.Ala635Val, and p.Ala650Asp — were identified across three unrelated families and shown to amplify JAK–STAT signaling downstream of interferon and IL-6.

Clinically, DPM is characterized by **rapidly progressive, circumferential, full-thickness sclerosis** that extends beyond the dermis into subcutaneous fat, fascia, muscle, and even bone, producing chronic non-healing skin ulcers, painful joint contractures, immobility, and a substantial risk of ulcer-related cutaneous squamous cell carcinoma (SCC). A distinguishing feature from systemic sclerosis is that **internal organ fibrosis is typically absent**. Mortality is high, driven by sepsis, gangrene, cardiopulmonary complications, and malignancy. A 2025 systematic review of 86 patients reported across a century (1923–2023) confirmed its rarity and poor prognosis, and notably found that the number of treatments administered did not influence outcome — underscoring the inadequacy of conventional immunosuppression ([PMID: 39520387](#)).

The identification of the STAT4-GOF/JAK-STAT mechanism transformed the therapeutic landscape. Whereas methotrexate, corticosteroids, and mycophenolate mofetil are frequently inadequate, the **JAK inhibitor ruxolitinib** reversed the hyperinflammatory fibroblast phenotype in vitro and produced resolution of inflammatory markers and clinical symptoms in treated patients without adverse effects. A 2025 independent Chinese case report replicated both the recurrent p.His623Tyr variant and the favorable ruxolitinib response, strengthening the genotype-guided treatment paradigm. DPM thus serves as a paradigm for how a rare "autoimmune"-appearing fibrosing disease can be a single-gene autoinflammatory disorder amenable to precision targeted therapy.

1. Disease Information

Overview. DPM is a rare, aggressive, childhood-onset fibrosing dermatosis. It sits at the severe extreme of the juvenile localized scleroderma (morphea) spectrum but is now understood to be genetically distinct in at least a subset of cases, caused by germline STAT4 GOF variants. The hallmark is deep, circumferential ("pansclerotic") sclerosis of the skin and underlying soft tissue, with ulceration, contractures, and progressive disability.

Key identifiers.

Resource	Identifier
OMIM (phenotype)	#620443 (DISABLING PANSCLEROTIC MORPHEA OF CHILDHOOD; DPMC)
MONDO	MONDO:0957497
Causal gene	STAT4 — OMIM *600558; NCBI Gene 6775; HGNC:11365; UniProt Q14765
Disease class	Juvenile localized scleroderma, pansclerotic subtype (Padua/PReS classification)

Synonyms / alternative names: Disabling pansclerotic morphea of childhood (DPMC); pansclerotic morphea; disabling pansclerotic morphea; (within JLS) pansclerotic/deep morphea subtype.

Nature of information. Evidence is derived primarily from **individual patient reports and small case series** aggregated into disease-level reviews, plus functional studies on patient-derived cells. The 2023 genetic discovery studied 4 patients from 3 families; the 2025 systematic review aggregated 86 patients from 52 reports.

2. Etiology

Primary cause (genetic). Heterozygous **gain-of-function missense variants in STAT4** (DNA-binding domain) cause autosomal dominant DPM ([PMID: 37256972](#)). *"Genome sequencing revealed three novel heterozygous missense gain-of-function variants in STAT4."* This is a **monogenic autoinflammatory etiology**, reclassifying the historically idiopathic/autoimmune disease.

Genetic risk factors. The causal variants are the GOF STAT4 DBD substitutions (p.His623Tyr, p.Ala635Val, p.Ala650Asp). No additional modifier loci have been formally mapped. Because variants are highly penetrant and dominant, carrying a single pathogenic allele is the principal risk determinant.

Environmental risk factors. In the broader JLS population, an environmental trigger (trauma, infection) is reported in ~13% of cases, and female sex predominates; however, for genetically-defined STAT4-DPM, the disease is driven by the germline variant. Within JLS generally, ANA positivity (42.3%) suggests an autoimmune diathesis, but this is not specific to DPM.

Protective factors. None established genetically or environmentally. (Not applicable / not reported.)

Gene–environment interactions. Not characterized for DPM. It is biologically plausible that interferon-inducing stimuli (viral infection) could exacerbate STAT4-GOF signaling, but this is inferred, not demonstrated.

3. Phenotypes

DPM phenotypes span cutaneous, musculoskeletal, hematologic/immunologic, and oncologic domains. Onset is in childhood (typically <14 years); course is **progressive**; severity is **severe**.

Phenotype	Type	HPO suggestion	Characteristics / frequency
Circumferential cutaneous sclerosis (to fascia/muscle/bone)	Physical manifestation	HP:0100324 (Skin sclerosis)	Hallmark; rapidly progressive; near-universal
Chronic skin ulcers	Clinical sign	HP:0100512 (Skin ulcer)	Common; prone to infection and SCC
Joint contractures	Clinical sign	HP:0001371 (Flexion contracture)	Common; cause immobility
Immobility / loss of ambulation	Functional	HP:0002540 (Inability to walk)	Progressive disability
Cutaneous squamous cell carcinoma (ulcer-related)	Neoplasm	HP:0002860 (Squamous cell carcinoma)	Recognized complication; can metastasize
Neutropenia	Lab abnormality	HP:0001875	Mild; in STAT4-DPM
Lymphopenia (CD4+ T cells)	Lab abnormality	HP:0001888 / HP:0005415	T-cell exhaustion, Th1 skew
Hypogammaglobulinemia (low IgG/IgA)	Lab abnormality	HP:0004313 / HP:0002720	Reduced IgG and IgA
Elevated inflammatory markers (ESR/CRP)	Lab abnormality	HP:0011227 (Elevated CRP)	Common
Peripheral eosinophilia	Lab abnormality	HP:0001880	In some patients

Distinguishing phenotype: *"Internal organ fibrosis is typically absent"* (PMID: 39520387) — a key separator from systemic sclerosis.

Quality of life. Profound: circumferential sclerosis, contractures, chronic ulcers, and pain cause severe impairment of mobility, self-care, and daily functioning, with lifelong disability and psychosocial burden. No disease-specific EQ-5D/SF-36 data are published; impact is inferred from the severe clinical course.

4. Genetic / Molecular Information

Causal gene. *STAT4* (Signal Transducer And Activator Of Transcription 4), OMIM *600558, located on chromosome 2q32.2–32.3.

Pathogenic variants (all heterozygous, germline, autosomal dominant; DNA-binding domain; transcript NM_003151.4):

Variant (cDNA)	Protein	Domain	Classification	Type	Consequence
c.1867C>T	p.His623Tyr (H623Y)	DBD	Pathogenic (GOF)	Missense	Gain of function
c.1904C>T	p.Ala635Val (A635V)	DBD	Pathogenic (GOF)	Missense	Gain of function
c.1949C>A	p.Ala650Asp (A650D)	DBD	Pathogenic (GOF)	Missense	Gain of function

These are **novel, private variants**, absent from population databases (gnomAD) — consistent with a severe, largely de novo dominant disorder. **Functional consequence: gain of function** — after interferon- α stimulation, variant-carrying cells showed increased/prolonged phospho-STAT4 relative to wild-type STAT4, and STAT4 is essential for transcriptional activation downstream of IL-6 receptor signaling ([PMID: 37256972](#)).

Somatic vs germline: Germline (constitutional); recurrent de novo events documented (p.His623Tyr seen in independent unrelated patients).

Modifier genes / epigenetics / chromosomal abnormalities: None established (not reported / not applicable).

5. Environmental Information

For genetically-defined STAT4-DPM, **the disease is fundamentally genetic** and no specific environmental toxin, pollutant, or occupational exposure is required or established. Within the broader morphea spectrum, trauma and infection are occasionally reported triggers (~13% of JLS), and interferon-inducing stimuli could theoretically amplify STAT4 signaling (inferred). **No infectious agent causes DPM.** Lifestyle factors are not implicated.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **heterozygous germline STAT4 DNA-binding-domain missense variant** (p.His623Tyr / p.Ala635Val / p.Ala650Asp) is present constitutionally → **leads to** an altered STAT4 transcription factor with increased activity.
2. Upon cytokine stimulation (interferon- α ; IL-12; IL-6-receptor signaling), the variant STAT4 **results in** increased and prolonged phosphorylation (hyperactive phospho-STAT4) relative to wild-type → i.e., **gain of function**.
3. Hyperactive STAT4 **leads to** amplified JAK–STAT transcriptional output and Th1-skewed, interferon-signature immune activation (demonstrated by single-cell RNA-seq immunodysregulatory signature).
4. This **results in** systemic immune dysregulation: mild neutropenia, CD4+ lymphopenia, T-cell exhaustion, and reduced IgG/IgA (hypogammaglobulinemia), with elevated inflammatory markers.
5. In parallel (branch), dermal **fibroblasts hypersecrete IL-6** and other inflammatory mediators → **leads to** a hyperinflammatory, profibrotic fibroblast state.
6. Dysfunctional fibroblasts show **impaired wound healing, impaired collagen-matrix contraction, and altered matrix secretion** → **results in** aberrant extracellular matrix deposition (fibrosis) plus poor re-epithelialization (chronic ulcers).
7. Deep, circumferential fibrosis **leads to** sclerosis extending to fascia, muscle, and bone, with contractures and immobility; chronic ulceration **leads to** infection risk and, over time, ulcer-related cutaneous SCC.

8. End results → disability, sepsis/gangrene, cardiopulmonary complications, malignancy, and high mortality.
9. **Therapeutic interruption:** JAK inhibition (ruxolitinib) blocks steps 2–3 (signal amplification) → reverses the hyperinflammatory fibroblast phenotype in vitro and resolves inflammatory markers and clinical symptoms in patients.

Supporting detail

- **Molecular pathways:** JAK–STAT signaling (type I/II interferon and IL-6/IL-12 axes); STAT4 is the key transducer. Upstream Janus kinases (JAK1/JAK2/TYK2) activate STAT4; downstream, excessive interferon-response and IL-6 transcriptional programs drive inflammation and fibrosis.
- **Cellular processes:** Chronic inflammation, Th1 polarization, T-cell exhaustion, impaired wound healing, and profibrotic fibroblast activation.
- **Protein dysfunction:** STAT4 DBD substitutions produce a **gain-of-function** transcription factor (increased/prolonged phosphorylation and transcriptional activity), not loss of function or aggregation.
- **Immune involvement:** Autoinflammatory (innate/adaptive dysregulation) rather than classic autoantibody-mediated autoimmunity; cytopenias and hypogammaglobulinemia indicate combined immune dysregulation.
- **Tissue damage mechanism:** Fibrosis (excess/aberrant collagen matrix) plus chronic ulceration; ischemia and infection compound tissue injury.
- **Molecular profiling:** Single-cell RNA-seq of patient PBMCs revealed an immunodysregulatory expression signature *"consistent with an immunodysregulatory phenotype that were appropriately modified through JAK inhibition."* Fibroblast assays quantified IL-6 hypersecretion and matrix dysfunction.

Suggested ontology terms: GO:0007259 (receptor signaling pathway via JAK–STAT); GO:0006954 (inflammatory response); GO:0042060 (wound healing); GO:0030198 (extracellular matrix organization). Cell types: CL:0000057 (fibroblast), CL:0000084 (T cell), CL:0000545 (Th1 cell), CL:0000775 (neutrophil). CHEBI: ruxolitinib (CHEBI:66919).

7. Anatomical Structures Affected

- **Organ/system level — primary:** Skin (UBERON:0002097) and skin of trunk/limbs; subcutaneous tissue; **musculoskeletal system** — fascia (UBERON:0007844), skeletal muscle (UBERON:0001134), joints (UBERON:0000982), and bone (UBERON:0001474).
 - **Secondary involvement:** Immune/hematologic system (cytopenias, hypogammaglobulinemia); potential cardiopulmonary complications contributing to mortality. **Internal visceral organ fibrosis is characteristically absent** (contrast with systemic sclerosis).
 - **Tissue types:** Connective tissue (dermal/subcutaneous collagen, fascia) is the principal target; epithelial (epidermal ulceration) and muscle tissue secondarily involved.
 - **Cell populations (CL):** Dermal fibroblasts (CL:0000057); T lymphocytes incl. CD4+/Th1 (CL:0000084, CL:0000545); neutrophils (CL:0000775); keratinocytes at ulcer margins (CL:0000312).
 - **Subcellular (GO Cellular Component):** Nucleus (GO:0005634; STAT4 acts as a nuclear transcription factor); cytoplasm (GO:0005737; latent STAT4 prior to activation).
 - **Localization / lateralization:** Typically **bilateral and circumferential** involvement of trunk and extremities; symmetric progression is common, though individual lesions may be asymmetric.
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8. Temporal Development

- **Onset:** Childhood, predominantly **<14 years of age** (juvenile-onset). Onset pattern is subacute-to-chronic with **rapid progression** of sclerosis.
- **Progression:** Relentlessly **progressive**; sclerosis rapidly becomes circumferential and deep (fascia/muscle/bone). Contractures and ulcers accumulate over time.
- **Disease course:** Chronic, lifelong, progressive; without effective mechanism-targeted therapy, outcomes are poor and mortality is high. The 2025 review found **female patients were younger at reported death**, and the **number of treatments did not influence outcome** under conventional management.
- **Remission:** Spontaneous remission is not characteristic. **Treatment-induced improvement** has now been demonstrated with ruxolitinib (resolution of inflammatory markers and symptoms).

- **Critical period:** Early intervention — ideally before irreversible deep fibrosis, contractures, and ulceration — represents the key window; genotype-guided JAK inhibition offers an opportunity to alter natural history.

9. Inheritance and Population

- **Epidemiology:** Ultra-rare. Only ~86 DPM patients have been reported in the literature (1923–2023; 52 reports). A Japanese nationwide survey estimated juvenile-onset morphea incidence at **2.11–2.87 per 1,000,000 children/year** (mean onset 7.7 yr); pansclerotic morphea is among the rarest, most severe subtypes requiring systemic immunosuppression. In the 750-patient international JLS cohort, deep morphea comprised only ~2% (linear 65%, plaque 26%, generalized 7%).
- **Inheritance: Autosomal dominant** (STAT4 GOF). Recurrent **de novo** variants documented (p.His623Tyr in unrelated patients).
- **Penetrance / expressivity:** High penetrance in reported families; expressivity appears severe. Formal penetrance estimates are limited by the small number of cases.
- **Genetic anticipation / mosaicism / founder effects:** Not established / not reported.
- **Consanguinity:** Not a factor (dominant mechanism).
- **Carrier frequency:** Variants are private and essentially absent from gnomAD; no meaningful population carrier frequency.
- **Demographics:** Affects children of both sexes; broader morphea shows female predominance. No specific ethnic predilection for STAT4-DPM (reported in multiple populations, including North American and Chinese patients). Sex ratio for STAT4-DPM is not firmly established given small numbers.

10. Diagnostics

Clinical diagnosis rests on the characteristic phenotype: rapidly progressive, circumferential, full-thickness sclerosis of trunk/limbs with ulceration and contractures.

Histopathology (skin biopsy): Deep dermal and subcutaneous hyalinized/sclerotic collagen, loss of adnexal structures, and thickened fascia/muscle involvement (deep morphea pattern).

Supportive laboratory findings: Cytopenias (neutropenia, CD4+ lymphopenia), hypogammaglobulinemia (low IgG/IgA), elevated ESR/CRP, and peripheral eosinophilia in some. ANA is frequently positive in localized scleroderma (42.3% in JLS) but **Scl-70 and anticentromere antibodies are usually negative** (helping exclude systemic sclerosis).

Imaging: MRI and ultrasound assess depth of fascial/muscle involvement and disease activity.

Genetic testing (confirmatory): Sequencing of *STAT4* (targeted single-gene, gene panel, exome, or genome) to identify heterozygous DNA-binding-domain GOF variants. Exome/genome sequencing was the discovery modality and is recommended for undiagnosed severe pansclerotic morphea.

Differential diagnosis:

Condition	Distinguishing feature from DPM
Systemic sclerosis	Internal-organ involvement, Raynaud, Scl-70+ (all absent in DPM)
Eosinophilic fasciitis	Can overlap/continuum; peripheral eosinophilia, fascial inflammation
Chronic graft-versus-host disease	History of transplant
Nephrogenic systemic fibrosis	Gadolinium exposure, renal failure
Scleromyxedema	Mucin deposition, paraproteinemia
Stiff skin syndrome	Congenital, non-inflammatory, <i>FBN1</i> -related
Other monogenic interferonopathies	Distinct genetic/clinical profiles

Screening: Cascade genetic testing of at-risk relatives once a familial *STAT4* variant is identified; no population newborn screening exists.

11. Outcome / Prognosis

- **Survival/mortality:** High mortality historically. Causes of death include **sepsis, gangrene, and cardiopulmonary involvement**; ulcer-related **metastatic SCC** also contributes. Female patients were younger at reported death.

- **Morbidity/disability:** Severe — progressive contractures, immobility, chronic ulceration, and pain lead to lifelong functional impairment.
- **Complications:** Chronic non-healing ulcers; secondary infection; cutaneous squamous cell carcinoma (can metastasize); joint contractures; cardiopulmonary involvement.
- **Recovery potential:** Deep fibrosis and contractures are largely irreversible; prevention of progression is the goal. Mechanism-targeted therapy (ruxolitinib) can halt inflammatory activity and improve symptoms.
- **Prognostic factors:** Depth/extent of sclerosis, presence of ulcers/SCC, and — critically — access to effective JAK-targeted therapy. Under conventional immunosuppression, **number of treatments did not influence outcome**, indicating a strong need for mechanism-based therapy.

12. Treatment

Conventional immunosuppression (often inadequate). For high-risk JLS including pansclerotic morphea, SHARE/EULAR consensus and juvenile scleroderma reviews recommend systemic therapy when disability is threatened:

Drug	Class	NCIT	Role
Methotrexate	Antimetabolite/DMARD	NCIT:C642	Cornerstone first-line
Systemic corticosteroids	Glucocorticoid	NCIT:C2271	Adjunct, induction
Mycophenolate mofetil	Immunosuppressant	NCIT:C2010	Severe/refractory

Despite these, the 86-patient review found **the number of treatments did not influence outcome**, underscoring poor efficacy of conventional immunosuppression in DPM.

Mechanism-targeted therapy — JAK inhibition. The 2023 STAT4-GOF discovery provided a rational target. **Ruxolitinib** (JAK1/2 inhibitor; NCIT:C82733) *"led to improvement in the hyperinflammatory fibroblast phenotype in vitro and resolution of inflammatory markers and clinical symptoms in treated patients, without adverse effects"* (PMID: 37256972). A 2025 independent case (6-year-old girl, de novo p.His623Tyr) achieved marked remission on corticosteroids plus ruxolitinib (5.0 mg AM + 2.5 mg PM) (PMID: 40518164).

Supportive and rehabilitative care. Aggressive wound care (ulcer prevention/healing), physiotherapy/occupational therapy to limit contractures, infection prophylaxis, pain management, and **SCC surveillance** of chronic ulcers.

Personalized medicine: Genotype-guided JAK–STAT blockade is the emerging standard once a STAT4 GOF variant is confirmed.

13. Prevention

- **Primary prevention:** Not possible for a de novo/dominant germline disease. Genetic counseling for affected families regarding recurrence risk (50% transmission from an affected parent; de novo risk otherwise).
- **Secondary prevention:** Early genetic diagnosis and prompt initiation of JAK-targeted therapy to prevent progression of fibrosis and contractures.
- **Tertiary prevention:** Prevent complications — vigilant ulcer/wound care, infection prophylaxis, contracture-limiting rehabilitation, and **surveillance/biopsy of chronic ulcers for SCC**.
- **Counseling:** Genetic counseling and cascade testing for at-risk relatives; prenatal/preimplantation testing is theoretically possible for a known familial variant.
- Immunization, public-health, and environmental interventions are not applicable to disease causation.

14. Other Species / Natural Disease

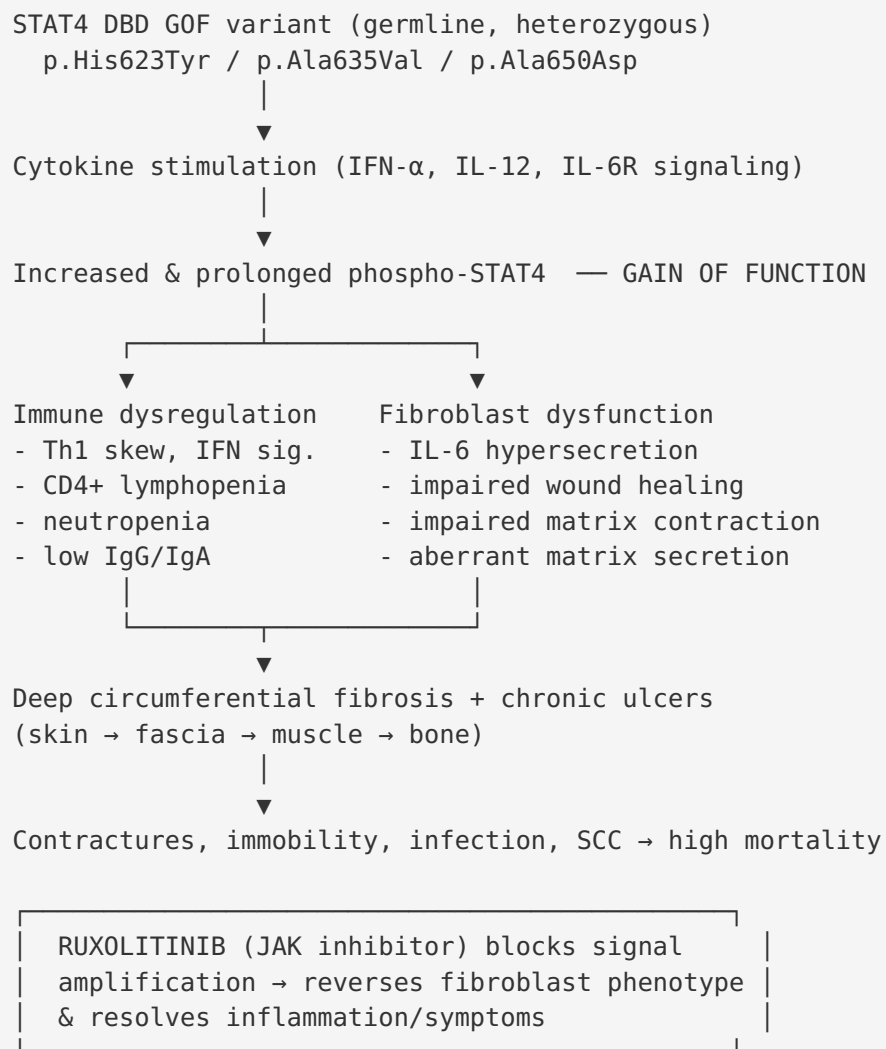
- **Taxonomy/orthologs:** *STAT4* is evolutionarily conserved. Human *STAT4* (NCBI Gene 6775; HGNC:11365; UniProt Q14765); mouse *Stat4* (NCBI Gene 20849); rat *Stat4* (NCBI Gene 362925).
- **Natural disease in animals:** No naturally occurring DPM-equivalent disease is catalogued in OMIA for companion animals or wildlife.
- **Comparative biology:** *Stat4*-knockout mice model **loss of function** (defective IL-12 signaling and impaired Th1 differentiation) and therefore do **not** represent this gain-of-function human disease.

- **Zoonotic/transmission:** Not applicable (non-infectious, genetic disease).
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15. Model Organisms

- **No dedicated animal model** specifically recapitulating STAT4-GOF DPM has been reported.
 - **Functional systems used:** (1) **Patient-derived primary dermal fibroblasts** demonstrating IL-6 hypersecretion, impaired wound healing, and impaired collagen-matrix contraction; (2) **cell lines reconstituted with wild-type vs variant STAT4** to measure phospho-STAT4 and transcriptional activity; (3) **single-cell RNA-seq of patient PBMCs** showing an immunodysregulatory signature normalized by JAK inhibition ([PMID: 37256972](#)). *"Primary skin fibroblast and cell-line assays were used to define the functional nature of the genetic defect."*
 - **Limitations:** Existing *Stat4*-knockout mice model the opposite (loss of function), so a **knock-in GOF mouse (e.g., Stat4 H623Y equivalent)** is needed to capture systemic and cutaneous disease in vivo.
 - **Applications:** Current cellular models support mechanistic dissection of JAK–STAT amplification, fibroblast dysfunction, and drug response (JAK-inhibitor screening).
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Mechanistic Model / Interpretation



DPM is best understood as a **single-gene autoinflammatory fibrosing disease**: an upstream STAT4 DNA-binding-domain gain-of-function lesion amplifies JAK–STAT signaling, which branches into systemic immune dysregulation and a cell-autonomous profibrotic, poorly-healing fibroblast state. The convergence of these branches produces the signature deep, circumferential sclerosis with ulceration. The upstream genetic/signaling node is pharmacologically "druggable" at the JAK level, which is why ruxolitinib — acting high in the causal chain — reverses downstream pathology where conventional broad immunosuppression (acting diffusely and non-specifically) fails.

Evidence Base

PMID	Study	Contribution
37256972	Baghdassarian et al., <i>NEJM</i> 2023	Landmark: identified 3 STAT4 DBD GOF variants in 3 families; defined fibroblast dysfunction & immunodysregulation; demonstrated ruxolitinib efficacy
39520387	Hua et al., <i>Br J Dermatol</i> 2025	Systematic review of 86 patients (1923–2023); clinical phenotype, absence of internal-organ fibrosis, mortality, treatment-outcome analysis
40518164	Shi et al., <i>Zhonghua Er Ke Za Zhi</i> 2025	Independent replication: de novo STAT4 p.His623Tyr with scleroderma, ruxolitinib response
16368732	Zulian et al., <i>Rheumatology</i> 2006	750-patient JLS cohort: subtype distribution, ANA 42.3%, environmental triggers
40040588	Japanese nationwide survey 2025	Juvenile morphea incidence (2.11–2.87 per 1,000,000/yr)
28543434	SCC in DPMC	Ulcer-related SCC complication
29455178	DPM mortality	Sepsis, gangrene, cardiopulmonary causes
24383741	DPM + eosinophilic fasciitis	Overlapping differential diagnosis

Key verbatim support: - *"Genome sequencing revealed three novel heterozygous missense gain-of-function variants in STAT4. In vitro, primary skin fibroblasts showed enhanced interleukin-6 secretion, with impaired wound healing, contraction of the collagen matrix, and matrix secretion."*

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- *"Inhibition of Janus kinase (JAK)–STAT signaling with ruxolitinib led to improvement in the hyperinflammatory fibroblast phenotype in vitro and resolution of inflammatory markers and clinical symptoms in treated patients, without adverse effects."*

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- *"Single-cell RNA sequencing revealed expression patterns consistent with an immunodysregulatory phenotype that were appropriately modified through JAK inhibition."*

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- "DPM is characterized by rapid sclerosis with circumferential involvement that frequently extends to the fascia, muscle and bone." and "Internal organ fibrosis is typically absent."

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- "Disabling pansclerotic morphea of childhood (DPMC) is a rare disorder that confers a risk of developing ulcer-related squamous cell carcinoma (SCC)."

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Limitations and Knowledge Gaps

1. **Small sample size.** Genetic evidence derives from only 4 patients in 3 families plus 1 replication case; penetrance, expressivity, and genotype–phenotype correlations remain imprecise.
2. **Genetic heterogeneity unresolved.** Not all historically-diagnosed DPM cases have been genotyped; some of the 86 reviewed patients may have non-STAT4 etiologies. The fraction of clinically-defined DPM that is STAT4-GOF is unknown.
3. **No in vivo model.** Absence of a GOF knock-in animal model limits study of systemic disease, long-term progression, and preclinical therapeutics.
4. **Durability and long-term safety of ruxolitinib** in children with DPM are not established in controlled trials; evidence is anecdotal/short-term.
5. **No disease-specific QoL, prevalence, or survival statistics** have been formally quantified; prognosis estimates are qualitative.
6. **Modifier genes, epigenetics, and gene–environment interactions** are entirely uncharacterized.

Proposed Follow-up Experiments / Actions

1. **Genotype legacy cohorts.** Sequence *STAT4* (and broader interferonopathy/fibrosis panels) in archived DPM cases to estimate the STAT4-GOF fraction and detect additional causal genes.

2. **Generate a Stat4-GOF knock-in mouse** (e.g., H623Y-equivalent) to recapitulate cutaneous/systemic disease and test JAK inhibitors in vivo.
 3. **Prospective registry + clinical trial of ruxolitinib** (and other JAK inhibitors/upstream biologics such as anti-IL-6) in genetically-confirmed patients, with standardized skin-score, imaging, and QoL endpoints.
 4. **Deep immunophenotyping and longitudinal single-cell/spatial transcriptomics** of skin and blood on and off therapy to define biomarkers of activity and response.
 5. **Structural/biochemical characterization** of DBD variants to explain how they increase STAT4 DNA binding/transcriptional output and to guide variant-specific therapeutics.
 6. **Establish SCC-surveillance and wound-care guidelines** specific to DPM given the malignancy risk.
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Consensus Answer

Disabling Pansclerotic Morphea of Childhood (OMIM #620443; MONDO:0957497) is an ultra-rare, childhood-onset, autosomal dominant autoinflammatory fibrosing disease caused by heterozygous gain-of-function missense variants in the DNA-binding domain of *STAT4* (p.His623Tyr, p.Ala635Val, p.Ala650Asp) that amplify JAK–STAT/IL-6/interferon signaling, producing hyperinflammatory, poorly-healing dermal fibroblasts, systemic immune dysregulation, and rapidly progressive circumferential sclerosis extending to fascia, muscle, and bone with ulcers, contractures, SCC risk, and high mortality while sparing internal organs. Conventional immunosuppression (methotrexate/corticosteroids/MMF) is largely ineffective, whereas the JAK inhibitor ruxolitinib reverses the cellular phenotype and improves patients, making genotype-guided JAK–STAT blockade the key mechanism-targeted therapy.
